

CALCULUS II

WEEK XI

FINDING EXTREME VALUES

I SECOND DERIVATIVE TEST

Suppose that $f(x,y)$ and its first and second partial derivatives are continuous throughout a disk centered at (a,b) and that

$f_x(a,b) = f_y(a,b) = 0$. Then,

$$D = D(a,b) = f_{xx}(a,b) \cdot f_{yy}(a,b) - f_{xy}^2(a,b)$$

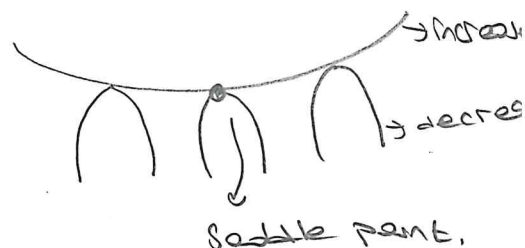
$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

Then,

- (1) If $D > 0$ and $\begin{cases} f_{xx}(a,b) > 0 \\ \text{or} \\ f_{yy}(a,b) > 0 \end{cases}$ then there is a local (relative) min at (a,b)
- (2) If $D > 0$ and $\begin{cases} f_{xx}(a,b) < 0 \\ \text{or} \\ f_{yy}(a,b) < 0 \end{cases}$ then there is a local (relative) max at (a,b)
- (3) If $D < 0$, then (a,b) is a saddle point (in every open

disk centered at (a,b) there are domain points where

$f(x,y) > f(a,b)$ and domain points (x,y) where $f(x,y) < f(a,b)$



- (4) $D = 0$ test is inconclusive.

Ex 7 Find and classify the critical points of $f(x,y) = xy e^{-(x^2+y^2)/2}$

$$\left. \begin{aligned} f_1(x,y) &= y(1-x^2) e^{-(x^2+y^2)/2} \\ f_2(x,y) &= x(1-y^2) e^{-(x^2+y^2)/2} \end{aligned} \right\} \text{critical points}$$

$(0,0), (1,1), (1,-1), (-1,1)$ and $(-1,-1)$

$$f_{11}(x,y) = xy(x^2-3) e^{-(x^2+y^2)/2}$$

$$f_{12}(x,y) = (1-x^2)(1-y^2) e^{-(x^2+y^2)/2}$$

$$f_{22}(x,y) = xy(y^2-3) e^{-(x^2+y^2)/2}$$

Let $(0,0)$ $f_{xx}(0,0) = 0$ $f_{xy}(0,0) = 1$ $f_{xx}(0,0)f_{yy}(0,0) - f_{xy}^2(0,0) = -1 < 0$
 $f_{yy}(0,0) = 0$ f is a saddle point.

$$f_{xx}(0,0)f_{yy}(0,0) - f_{xy}^2(0,0) = -1 < 0$$

$(1,1)$ & $(-1,-1)$ $f_{xx}(1,1) = f_{xx}(-1,-1) = f_{yy}(1,1) = f_{yy}(-1,-1) = -2/e < 0$

$$f_{xy}(1,1) = f_{xy}(-1,-1) = 0$$

we have a local max.

$$f_{xx}(1,1)f_{yy}(1,1) - f_{xy}^2(1,1) = f_{xx}(-1,-1)f_{yy}(-1,-1) - f_{xy}^2(-1,-1) = 4/e$$

$(1,-1)$ & $(-1,1)$ $f_{xx}(1,-1) = f_{xx}(-1,1) = 2/e > 0$ $f_{xy}(1,-1) = f_{xy}(-1,1) = 0$
 $f_{yy}(1,-1) = f_{yy}(-1,1) = 2/e > 0$

we have a local min at that point

$$0 < \frac{4}{e^2} = f_{xx}(1,-1)f_{yy}(1,-1) - f_{xy}^2(1,-1) = f_{xx}(-1,1)f_{yy}(-1,1) - f_{xy}^2(-1,1) =$$

Ex 7 Find and classify all the critical points of $f(x,y) = 4 + x^3 + y^3 - 3x$

$$f_x = 3x^2 - 3 = 0 \Rightarrow x^2 = 1$$

$$f_y = 3y^2 - 3 = 0 \Rightarrow y^2 = 1$$

$$3x^2 - 3 = 0 \Rightarrow 3x(x-1) = 0$$

$$\Downarrow$$

$$x=0 \Rightarrow y=0 \quad (0,0)$$

$$x=1 \Rightarrow y=1 \quad (1,1)$$

Hence $(0,0)$ and $(1,1)$ are critical points

$$f_{xx}(x,y) = 6x \quad f_{yy}(x,y) = 6y \quad f_{xy} = 0$$

$$D(x,y) = f_{xx}(x,y)f_{yy}(x,y) - f_{xy}^2(x,y) = 36xy - 0$$

$(0,0)$ $D(0,0) = -9 < 0$ $(0,0)$ is a saddle point

$(1,1)$ $D(1,1) = 36 > 0$ $f_{xx}(1,1) = 6 > 0$ local min point is $(1,1)$
minimum value is $f(1,1) = 3$

Ex Determine the part on the plane $4x - 2y + z = 1$ that is closest to the point $(-2, -1, 5)$

The distance between the desired point and $(-2, -1, 5)$ is given by

$$d = \sqrt{(x+2)^2 + (y+1)^2 + (z-5)^2}$$

Since $z = 1 - 4x + 2y$, we have

$$d(x,y) = \sqrt{(x+2)^2 + (y+1)^2 + (-4x+2y-4)^2}$$

Finding the minimum value of $d(x,y)$ is equivalent to

finding the minimum value of $d^2(x,y)$

So consider the function

$$f(x,y) = d^2(x,y) = (x+2)^2 + (y+1)^2 + (-4x+2y-4)^2$$

$$f_x(x,y) = 2(x+2) - 8(-4x+2y-4) = 34x - 16y + 36 = 0$$

$$f_y(x,y) = 2(y+1) + 4(-4x+2y-4) = 10y - 16x - 14 = 0$$

$$f_{xx} = 34 \quad f_{yy} = 10 \quad f_{xy} = -16 \Rightarrow D = 34 \cdot 10 - (-16)^2 = 84 > 0$$

(3)

is the only critical point.

Since $D > 0$ and $f_{xx} > 0$, we know that this critical point is a local minimum. The z -coordinate is

$$z = 1 - 4\left(-\frac{34}{21}\right) + 2\left(-\frac{25}{21}\right) = \frac{107}{21}$$

So, the point closest to $(-2, 1, 3)$ is $\left(-\frac{34}{21}, -\frac{25}{21}, \frac{107}{21}\right)$ on the plane

ABSOLUTE MAXIMUM AND MINIMUM VALUES

Definitions

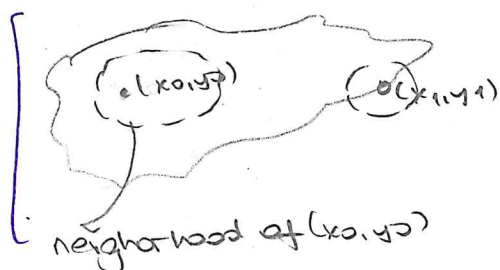
- A region in $\mathbb{R}^2$ is called **closed** if it includes its boundary. A region is called **open** if it doesn't include any of its boundary points.
- A region in $\mathbb{R}^2$ is called **bounded** if it can be completely contained in a disk. In other words, a region will be bounded if it is finite.



Closed



Sets that are not closed



- (x_0, y_0) interior point of region R , that is, neighborhood of $(x_0, y_0) \subset R$
- (x_1, y_1) boundary point of region R , neighborhood of $(x_1, y_1) \not\subset R$

THE EXTREME VALUE THEOREM If $f(x, y)$ continuous function of n variables whose domain is closed and bounded in $\mathbb{R}^n$, then the range of f is a bounded set of real numbers, and there are points in its domain where f takes on absolute maximum and minimum values

II FINDING ABSOLUTE EXTREMA ON CLOSED BOUNDED REGION

Consider $f(x,y)$ is continuous on a closed bounded region D

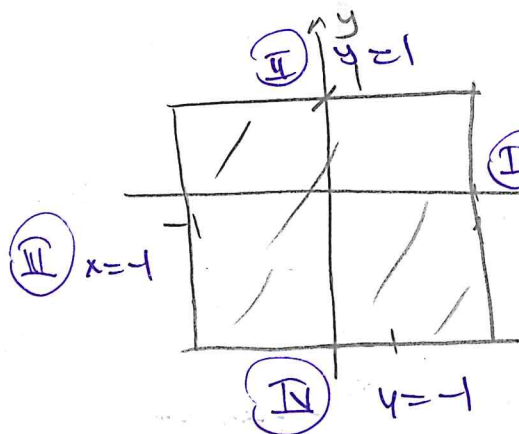
① INTERIOR POINTS Find all the critical points of the function that lie in the region D and determine the function value at each of these points.

② BOUNDARY POINTS Find all extrema of the function on the boundary.

③ The LARGEST and SMALLEST VALUES found in the first two steps are ABSOLUTE MINIMUM and ABSOLUTE MAXIMUM of the function.

Ex Find the absolute minimum and absolute maximum of

$f(x,y) = x^2 + 4y^2 - 2x^2y + 4$ on the rectangle given by $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$



INTERIOR POINT (CRITICAL POINTS)

$$f(x,y) = x^2 + 4y^2 - 2x^2y + 4$$

$$f_x = 2x - 4xy = 0$$

$$f_y = 8y - 2x^2 = 0 \Rightarrow y = x^2/4$$

$$2x - x^3 = 0 \Rightarrow x = 0 \quad x = \pm\sqrt{2} \quad \text{not in the region}$$

Since $y = x^2/4$, $y = 0$ we have region.
 $(0,0) \quad \underline{f(0,0) = 4}$

BOUNDARY POINTS ① $x=1$ $-1 \leq y \leq 1$

$$g(y) = f(1,y) = 1 + 4y^2 - 2y + 4 = 4y^2 - 2y + 5$$

$$g'(y) = 8y - 2 = 0 \quad y = 1/4 \quad (1, 1/4) \text{ critical point}$$

So, we have points $(1, -1), (1, 1), (1, 1/4)$

$$\underline{f(1, -1) = 11 = g(-1)}$$

$$\underline{f(1, 1) = 7 = g(1)}$$

$$\underline{f(1, 1/4) = 19/4 = g(1/4)}$$

(5)

II $y=1 \quad -1 \leq x \leq 1$

$$h(x) = f(x, 1) = x^2 + 4 - 2x^2 + 4 = 8 - x^2$$

$$h'(x) = -2x = 0 \Rightarrow x=0 \text{ critical point } (0, 1)$$

We have points $(-1, 1), (1, 1), (0, 1)$

$$\boxed{h(-1) = f(-1, 1) = 7}$$

$$\boxed{h(1) = f(1, 1) = 7}$$

$$\boxed{h(0) = f(0, 1) = 8}$$

III $x=-1 \quad -1 \leq y \leq 1$

$$g(y) = f(-1, y) = 4y^2 - 2y + 5 \quad g'(y) = 8y - 2 \Rightarrow y = 1/4 \text{ critical point}$$

We have points $(-1, 1/4), (-1, -1), (-1, 1)$

$$\boxed{g(-1) = f(-1, -1) = 11}$$

$$\boxed{g(1) = f(-1, 1) = 7}$$

$$\boxed{g(1/4) = f(-1, 1/4) = 19/4}$$

IV $y=-1 \quad -1 \leq x \leq 1$

$$h(x) = f(x, -1) = x^2 + 4 + 2x^2 + 4 = 3x^2 + 8$$

$$h'(x) = 6x = 0 \Rightarrow x=0 \text{ critical point.}$$

$$\boxed{h(-1) = f(-1, -1) = 11}$$

$$\boxed{h(1) = f(1, -1) = 11}$$

$$\boxed{h(0) = f(0, -1) = 8}$$

We have

$$\boxed{f(0, 0) = 4} \rightarrow \text{ABS MIN}$$

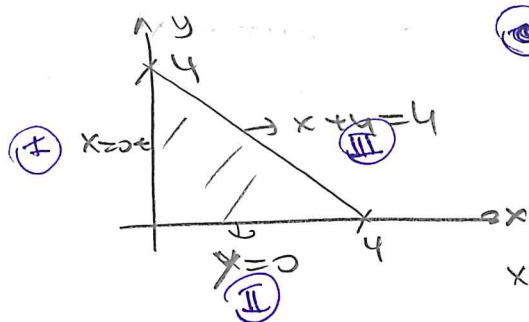
$$\boxed{f(1, 1/4) = f(-1, 1/4) = 19/4}$$

$$\boxed{f(1, -1) = f(-1, -1) = 11} \rightarrow \text{ABS MAX}$$

$$\boxed{f(-1, 1) = f(1, 1) = 7}$$

$$\boxed{f(0, 1) = f(0, -1) = 8}$$

Ex 7 Find abs max and min of the function $f(x,y) = x^2 y e^{-(x+y)}$ in the closed and bounded region $x \geq 0, y \geq 0, x+y \leq 4$



① INTERIOR POINTS

$$f_x(x,y) = x y (2-x) e^{-(x+y)}$$

$$f_y(x,y) = x^2 (1-y) e^{-(x+y)}$$

$$f_x(x,y) = 0 = f_y(x,y)$$

$$x y (2-x) = 0 \Rightarrow x=0, y=0, x=2$$

$$x^2 (1-y) = 0 \Rightarrow x=0, y=1, (0,1)$$

$$y=0, x=0, (0,0)$$

$$x=2, y=1, (2,1)$$

We have three points $(0,1), (0,0), (2,1)$. Since $(0,0)$ and $(0,1)$ are boundary points we can analyze them later.

$$f(2,1) = 4/e^3 \approx 0.199$$

② BOUNDARY POINTS

(I) $x=0, f(0,y)=0$ } Boundary points $(0,0)$ and $(0,1)$ are here.

(II) $y=0, f(x,0)=0$

(III) $y=4-x, f(x,4-x) = x^2(4-x)e^{-(x+4-x)} = x^2(4-x)e^{-4} = p(x)$

$$f'(x,4-x) = p'(x) = (8x-3x^2)e^{-4}$$

$$f'(x,4-x) = 0 \Rightarrow x(8-3x)e^{-4} = 0 \Rightarrow x=0 \quad x=8/3$$

$$\Downarrow \quad \Downarrow$$

$$y=4 \quad y=4/3$$

So, we have boundary points $(0,4), (8/3, 4/3)$

$$f(0,4) = 0, f(8/3, 4/3) = \frac{256}{27} e^{-4} \approx 0.174$$

Absolute maximum at point $(2,1)$, abs max value is $f(2,1) \approx 0.199$

Abs min at $(0,y), (x,0)$, abs min is $f(0,y) = f(x,0) = 0$

Ex 7 Find the maximum and minimum values of $f(x,y) = 2xy$ on the closed disk $x^2 + y^2 \leq 4$.

Since $x^2 + y^2 \leq 4$ is closed and $f(x,y) = 2xy$ is continuous, f must have absolute maximum and minimum values at some points of the disk.

$$f_x(x,y) = 2y \quad f_y(x,y) = 2x$$

INT Only critical value is $(0,0)$ $f(0,0) = 0$.

BOUNDARY $x = 2\cos t$, $y = 2\sin t$ $-\pi \leq t \leq \pi$

$$f(2\cos t, 2\sin t) = 8\cos t \cdot \sin t = g(t)$$

$$0 = g'(t) = -8\sin^2 t + 8\cos^2 t \quad (\Rightarrow) \quad \tan^2 t = 1 \quad (\Rightarrow) \quad t = \pm \pi/4 \text{ or } \pm 3\pi/4$$

$$g(\pi/4) = 8 \cdot \cos(\pi/4) \cdot \sin(\pi/4) = 8 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} = 4$$

$$g(-3\pi/4) = 8 \cos(3\pi/4) \sin(-\pi/4) = -4$$

$$t = \pi/4 \Rightarrow (x,y) = (\sqrt{2}, \sqrt{2}) \quad (x,y) = (-\sqrt{2}, -\sqrt{2})$$

$$t = -3\pi/4 \Rightarrow (x,y) = (\sqrt{2}, -\sqrt{2}) \quad (x,y) = (-\sqrt{2}, \sqrt{2})$$

① f has absolute max value 4 at $(\sqrt{2}, \sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$
and absolute min value -4 at $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$

III LAGRANGE MULTIPLIERS

Objective Dealing with problems of maximizing or minimizing functions whose variables are not independent of one another.

Thm Suppose that f and g have continuous first partial derivatives near the point $P_0 = (x_0, y_0)$ on the curve C with equation $g(x, y) = 0$. Suppose also that, when restricted to points on C , the function $f(x, y)$ has a local maximum or minimum value at P_0 . Finally, suppose that

① P_0 is not an endpoint of C , and

② $\nabla g(P_0) \neq 0$

Then, there exists a number λ_0 such that (x_0, y_0, λ_0) is the critical point of the Lagrangian function.

$$L(x, y) = f(x, y) + \lambda g(x, y)$$

— Consider the next to optimize the function

$f(x, y, z)$ subject to the constraint $g(x, y, z) = k$

→ To find candidates for points on curve $g(x, y) = 0$ at $f(x, y)$ is maximum or minimum, we should look for critical points of the Lagrangian function

$$L(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

At any critical point of L we must have

$$\left. \begin{aligned} 0 &= \frac{\partial L}{\partial x} = f_x(x, y) + \lambda g_x(x, y) \\ 0 &= \frac{\partial L}{\partial y} = f_y(x, y) + \lambda g_y(x, y) \end{aligned} \right\} \text{that is } \nabla f \parallel \nabla g$$

$$0 = \frac{\partial L}{\partial \lambda} = g(x, y)$$

→ the constraint equation.

★ NOTE IT IS ASSUMED THAT THE CONSTRAINED PROBLEM

Ex Find the shortest distance from the origin to the curve $x^2y = 16$

Question $\Rightarrow$ minimize $f(x,y) = \sqrt{x^2+y^2}$ (distance from origin)
 Subject to $g(x,y) = x^2y - 16 = 0$

$$L(x,y) = \underbrace{x^2+y^2}_{f(x,y)} + \lambda \underbrace{(x^2y-16)}_{g(x,y)}$$

not on the curve

$$0 = \frac{\partial L}{\partial x} = 2x + 2\lambda xy = 2x(1 + \lambda y) \Rightarrow \boxed{x=0} \text{ or } \lambda y = -1$$

$$0 = \frac{\partial L}{\partial y} = 2y + \lambda x^2$$

multiply with y

$$2y^2 + \lambda x^2y = 2y^2 - x^2 = 0$$

$$\lambda y = -1$$

$$0 = \frac{\partial L}{\partial \lambda} = x^2y - 16$$

$\rightarrow$

$$x^2y - 16 = 0 \quad 2y^2 - x^2 = 0$$

$$\underbrace{x^2y - 16 = 0 \quad 2y^2 - x^2 = 0}_{x^2 = 2y^2}$$

$$2y^3 = 16 \quad y = 2$$

Since our point will satisfy $x^2y = 16$ for $y=2$ $x = \pm 2\sqrt{2}$

then we have points $(\pm 2\sqrt{2}, 2)$

the minimum distance $= 2\sqrt{3}$

Ex Find the points on the curve $17x^2 + 12xy + 8y^2 = 100$ that are closest and farthest away from the origin.

$$17x^2 + 12xy + 8y^2 = 8x^2 + (3x+2y)^2 + 4y^2 \text{ is positive definite.}$$

Thus, $17x^2 + 12xy + 8y^2 = 100$ is a bounded curve

$$f(x,y) = x^2 + y^2$$

$$g(x,y) = 17x^2 + 12xy + 8y^2 = 100$$

$$L(x,y) = x^2 + y^2 + \lambda(17x^2 + 12xy + 8y^2 - 100)$$

$$0 = \frac{\partial L}{\partial x} = 2x + \lambda(34x + 12y)$$

$$0 = \frac{\partial L}{\partial y} = 2y + \lambda(12x + 16y)$$

$$0 = \frac{\partial L}{\partial \lambda} = 17x^2 + 12xy + 8y^2 - 100$$

$$\frac{-2x}{34x+12y} = \frac{-2y}{12x+16y}$$

$$-24x^2 - 32xy = -68xy - 24y^2$$

$$-24x^2 + 36xy + 24y^2 = 0$$

$$-2x^2 + 3xy + 2y^2 = 0$$

$$2x^2 - 3xy - 2y^2 = 0$$

$$17x^2 + 12xy + 8y^2 - 100 = 0$$

$$17x^2 + 8x^2 - 8y^2 + 8y^2 - 100 = 0$$

$$25x^2 = 100$$

$$x = \pm 2$$

$$2x^2 - 3xy - 2y^2 = 0$$

$$3xy = 2x^2 - 2y^2$$

$$12xy = 8x^2 - 8y^2$$

$$\begin{aligned} x &= 2 \\ y^2 + 3y - 4 &= 0 \\ (y-1)(y+4) &= 0 \end{aligned}$$

$$\begin{aligned} x &= -2 \\ y^2 - 3y - 4 &= 0 \\ (y+1)(y-4) &= 0 \end{aligned}$$

Candidate points

$(2,1)$, $(-2,-1)$, $(2,-4)$ and $(-2,4)$

$$f(x,y) = 5$$

MIN.

$$f(x,y) = 20$$

MAX.

Ex Find the dimensions of the box with largest volume if the total surface area is 64 cm^2



$$f(x, y, z) = V(x, y, z) = xyz \quad \text{volume of the box}$$

Total surface area

$$2xy + 2yz + 2xz = 64$$

$$g(x, y, z) = xy + yz + xz - 32 = 0$$

$$L(x, y, z) = f(x, y, z) + \lambda g(x, y, z)$$

$$L(x, y, z) = xyz + \lambda (xy + yz + xz - 32)$$

$$\frac{\partial L}{\partial x} = yz + \lambda(y + z) = 0$$

$$\frac{\partial L}{\partial y} = xz + \lambda(x + z) = 0$$

$$\frac{\partial L}{\partial z} = yx + \lambda(y + x) = 0$$

$$\frac{\partial L}{\partial \lambda} = xy + yz + xz - 32 = 0$$

$$\cancel{xy} + \lambda x(y+z) = \cancel{xyz} + \lambda y(x+z) \quad \text{--- } \frac{\partial L}{\partial x} = 0$$

$$\lambda x(y+z) = \lambda y(x+z) \quad \text{--- } \textcircled{1}$$

$$\cancel{xy} + \lambda y(x+z) = \cancel{xyz} + \lambda z(y+x) \quad \text{--- } \frac{\partial L}{\partial y} = 0$$

$$\lambda y(x+z) = \lambda z(y+x) \quad \text{--- } \textcircled{2}$$

From $\textcircled{1}$ we have $\lambda(xz - yz) = 0 \Rightarrow \lambda = 0$ or $xz = yz$

Case I $\lambda = 0$ is not possible, since it gives the case

we would have from $f(x) + \lambda g(x) = 0 \Rightarrow yz = 0 \Rightarrow y = 0$ or $z = 0$ (not possible) if $x > 0$

Case II $xz = yz \Rightarrow x = y$ now check what is z .

we can get it from $\textcircled{2}$

$$xy(x+z) = xz(y+x)$$

$$yx + yz = yz + zx$$

$$yx = zx \Rightarrow y = z$$

Which means we have

$$x = y = z$$

Using the constraint $g(x, y, z) = xy + yz + xz = 32$ we have

$$3x^2 = 32 \quad x = \pm \sqrt{\frac{32}{3}} \quad x \text{ is a length so we have}$$

$$x = y = z = \frac{\sqrt{32}}{\sqrt{3}}$$

Ex 7 Find the maximum and minimum values of the function $f(x,y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$. $g(x,y) = x^2 + y^2 - 1 = 0$
constraint

$$\underbrace{f(x,y) = 3x + 4y}_{\text{function.}}$$

We will solve $L = f + \lambda g$ and $g(x,y) = x^2 + y^2 - 1 = 0$ together.

$$L(x,y) = (3x + 4y) + \lambda (x^2 + y^2 - 1)$$

$$\frac{\partial L}{\partial x} = 3 + 2\lambda x = 0$$

$$x = -3/2\lambda$$

$$\frac{\partial L}{\partial y} = 4 + 2\lambda y = 0$$

$$y = -4/2\lambda$$

$$\left\{ \begin{array}{l} x^2 + y^2 - 1 = 0 \\ \frac{9}{4\lambda^2} + \frac{16}{4\lambda^2} - 1 = 0 \end{array} \right.$$

$$\frac{25}{4\lambda^2} = 1$$

$$\lambda = \pm 5/2$$

$$\frac{\partial L}{\partial \lambda} = x^2 + y^2 - 1 = 0$$

$$\lambda = \pm 5/2$$

$$x = \frac{3}{2\lambda} = \mp \frac{3}{5}$$

$$y = \frac{2}{\lambda} = \mp \frac{4}{5}$$

$f(x,y)$ has extreme values at $(\mp \frac{3}{5}, \mp \frac{4}{5})$

$$\underline{\text{Max}} \quad 3\left(\frac{3}{5}\right) + \frac{4}{5} \cdot 4 = 5$$

$$\underline{\text{Min}} \quad 3\left(-\frac{3}{5}\right) + 4\left(-\frac{4}{5}\right) = -5$$

Ex 7 Find the maximum and minimum values of $f(x,y) = 4x^2 + 10y^2$ on the disk $x^2 + y^2 \leq 4$.

The constraint is the inequality for the disk. This is a closed bounded region, by the Extreme Value Theorem we know that a maximum and minimum value must exist.

First find all critical points that are in the disk

$$f_x = 8x = 0 \Rightarrow x = 0$$

$$\Rightarrow x = 0$$

(0,0) is the only point

$$f_y = 20y = 0 \Rightarrow y = 0$$

$$\Rightarrow y = 0$$

and it is in the disk.

Therefore, we can check only point at the boundaries and this means that we will check points on $x^2 + y^2 = 4$

$$g(x,y) = x^2 + y^2 - 4 = 0 \quad f(x,y) = 4x^2 + 10y^2$$

$$L(x,y) = 4x^2 + 10y^2 + \lambda(x^2 + y^2 - 4)$$

$$\frac{\partial L}{\partial x} = 8x + 2\lambda = 0 \Rightarrow 2x(4 - \lambda) = 0 \quad x=0 \text{ or } \lambda=4$$

$$\frac{\partial L}{\partial y} = 20y + 2\lambda y = 0 \Rightarrow 2y(10 - \lambda) = 0 \quad y=0 \text{ or } \lambda=10$$

$$\frac{\partial L}{\partial \lambda} = (x^2 + y^2 - 4) = 0$$

$$x=0 \text{ gives } \underbrace{y^2 = 4}_{x^2 + y^2 = 4} \quad (0, 2), (0, -2)$$

$$y=0 \text{ gives } x = \pm 2 \quad (2, 0), (-2, 0)$$

$$f(0, 0) = 0 \quad \text{Minimum}$$

$$f(2, 0) = f(-2, 0) = 16$$

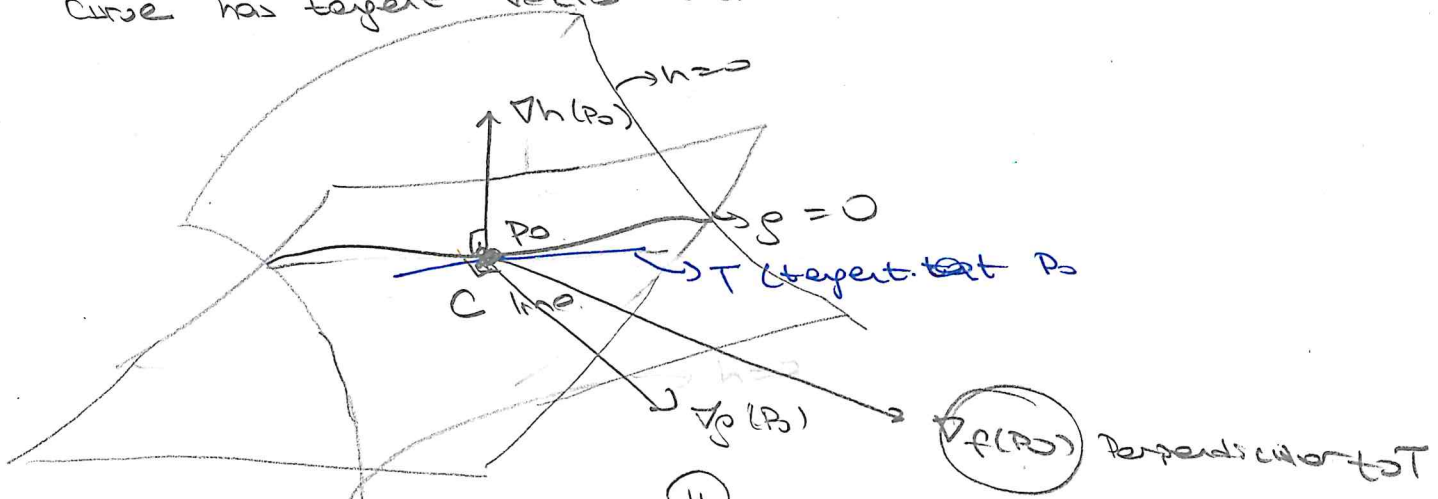
$$f(0, 2) = f(0, -2) = 40 \quad \text{Maximum}$$

PROBLEMS MORE THAN ONE CONSTRAINT

Extremize $f(x,y,z)$ subject to $g(x,y,z) = 0$ and $h(x,y,z) = 0$.

(Lagrange's theorem, constraint)
 Again, we assume that the problem has a solution, say, at point $P_0 = (x_0, y_0, z_0)$ and the function f, g, h have continuous first partial derivatives near P_0 .

Also we assume $\vec{T} = \nabla g(P_0) \times \nabla h(P_0) \neq 0$. These conditions imply that the surfaces $g(x,y,z) = 0$ and $h(x,y,z) = 0$ are smooth near P_0 and are not tangent to each other there, so they must intersect in a curve C that is smooth near P_0 and are not tangent to each other there so they must intersect in a curve C that is smooth near P_0 . The curve has tangent vector T at P_0 .



Since $\nabla g(P_0)$ and $\nabla h(P_0)$ are nonzero and both perpendicular to $T \nabla f(P_0)$ must lie in the plane spanned by these two vectors and must be a linear combination of them

$$\nabla f(x_0, y_0, z_0) = -\lambda_0 \nabla g(x_0, y_0, z_0) - \mu_0 \nabla h(x_0, y_0, z_0)$$

for some λ_0, μ_0 . It follows that $(x_0, y_0, z_0, \lambda_0, \mu_0)$ is a critical point of the Lagrangian function $L(x, y, z, \lambda, \mu) = f + \lambda g + \mu h$

we look for (x, y, z) that extremize $f(x, y, z)$ subject to the two constraints $g(x, y, z) = 0$ and $h(x, y, z) = 0$

$$f_1 + \lambda g_1 + \mu h_1 = 0$$

$$f_2 + \lambda g_2 + \mu h_2 = 0$$

$$f_3 + \lambda g_3 + \mu h_3 = 0$$

$$g = 0$$

$$h = 0$$

Ex Find max and min values of $f(x, y, z) = xy + 2z$ on the circle that is the intersection of the plane $x + y + z = 0$ and the sphere $x^2 + y^2 + z^2 = 24$

$$\text{So } f = xy + 2z$$

$$g = x + y + z = 0$$

$$h = x^2 + y^2 + z^2 - 24 = 0$$

$$L = xy + 2z + \lambda(x + y + z) + \mu(x^2 + y^2 + z^2 - 24)$$

$$\textcircled{I} \quad \frac{\partial L}{\partial x} = y + \lambda + 2\mu x = 0 \quad \left\{ \begin{array}{l} (x - y)(1 - 2\mu) = 0 \\ \mu = 1/2 \text{ or } x = y \end{array} \right.$$

$$\textcircled{II} \quad \frac{\partial L}{\partial y} = x + \lambda + 2\mu y = 0$$

$$\textcircled{III} \quad \frac{\partial L}{\partial z} = 2 + \lambda + 2\mu z = 0$$

$$\textcircled{IV} \quad \frac{\partial L}{\partial \lambda} = x + y + z = 0$$

$$\textcircled{V} \quad \frac{\partial L}{\partial \mu} = x^2 + y^2 + z^2 - 24 = 0$$

CASE I $\mu = 1/2$

$$\text{From } \textcircled{II} \quad x + \mu + y = 0 \quad \left\{ \begin{array}{l} x + y = -2 - \mu \\ \text{From } \textcircled{III} \quad 2 + \lambda + z = 0 \end{array} \right.$$

$$\text{From } \textcircled{IV} \quad z = -1 \quad x + y = 1$$

$$\text{From } \textcircled{V} \quad x^2 + y^2 = 24 - z^2 = 23$$

$$(x + y)^2 = x^2 + y^2 + 2xy = 1$$

$$\textcircled{VI} \quad 23 \quad 2xy = -22$$

$$2xy = -22 \quad xy = -11 \quad (x-y)^2 = x^2 + y^2 - 2xy = 23 + 22 = 45$$

$$(x-y) = \pm 3\sqrt{5}$$

we have $x-y = \pm 3\sqrt{5}$ so $\mu = 1/2$ we have points
 $x+y = 1$

$$(x_1, y_1, z_1) = \left(\frac{1+3\sqrt{5}}{2}, \frac{1-3\sqrt{5}}{2}, -1 \right)$$

$$(x_2, y_2, z_2) = \left(\frac{1-3\sqrt{5}}{2}, \frac{1+3\sqrt{5}}{2}, -1 \right)$$

$$f(x_1, y_1, z_1) = f(x_2, y_2, z_2) = -11 - 2 = -13$$

CASE II

$$x=y$$

since $x+y+z=0$ $z = -2x$

Since $x^2 + y^2 + z^2 - 24 = 0$ $x^2 + x^2 + 4x^2 - 24 = 0$
 $6x^2 = 24 \quad x = \pm 2$

Therefore, points $(2, 2, -4)$ and $(-2, -2, 4)$ must be considered

$$f(2, 2, -4) = 4 - 8 = -4$$

$$f(-2, -2, 4) = 4 + 8 = 12 \quad \text{MAXIMUM}$$

AND FROM CASE I

$$f(x_1, y_1, z_1) = f(x_2, y_2, z_2) = -13 \quad \text{MINIMUM}$$

PROBLEMS MORE THAN ONE CONSTRAINT

Extreme $f(x) = 0 \quad x = (x_1, x_2, \dots, x_n)$

Subject to $g_1(x) = 0, \dots, g_m(x) = 0$

$$L(x, \lambda_1, \lambda_2, \dots, \lambda_m) = f(x) + \sum_{j=1}^m \lambda_j^0 g_j^0(x)$$

Any critical points must satisfy the $n+m$ equations

$$\frac{\partial L}{\partial x_i} = 0 \quad (1 \leq i \leq n) \quad \frac{\partial L}{\partial \lambda_j^0} = g_j^0(x) = 0 \quad 1 \leq j \leq m$$